Locomotive Valve and Cylinder Lubrication

By W. J. Schlacks

In the beginning of locomotive operation the valves and cylinders were lubricated with chunks of tallow put into containers called "tallow pots" located in the top of the steam chest covers. When cylinder oil was first introduced, it was put into similar containers in the same location. It was difficult to control the amount of lubricant delivered, so that it was a case of either "a feast or a famine," in that at times entirely too much lubricant was fed into the steam until the supply was exhausted after which the valves and cylinders would run dry. It was a very uncertain affair; one never knew when starting out just how long his lubricant would last. Oil cups with pipes leading to each steam chest were later placed in the cab, thus ensuring more frequent filling and avoiding the necessity of the enginemen's risking injury in filling the tallow pots out in front while running.

The introduction of the hydrostatic lubricator filled a long felt want, because it could be located in the cab convenient for observation by the engineer. It could be regulated to feed a certain number drops per minute that ensured a certain consumption, more or less equally divided, for the entire trip. It held its position as a standard of locomotive equipment for over fifty years.

With the advent of the superheater locomotive, however, came demands for more exacting oil delivery. The increased length of the locomotives with the increased height of the point of oil delivery into the steam, on account of the larger drivers and cylinders with the piston valve above, together with the fact that the lubricator in the cab could not be raised to give the same incline to the oil pipes on account of the restricted overhead clearance, combined to put the hydrostatic lubricator at a disadvantage as a regular oil delivering instrument, by reducing the oil pipe pressure at the point of injecting the oil into the steam, thus making dependable regular delivery into the steam, against a full open throttle pressure, impossible.

The constantly increasing number of devices applied to locomotives for reducing the cost of operation, together with the constantly increasing number of signals and automobile road crossings that he must whistle, are consuming so much of the engineer's time, that it is being found increasingly difficult to demand his attention to frequently regulate, at so many drops per minute, the oil fed to the valves and cylinders, so as to deliver a sufficient amount, and not too much, at different speeds and cuts-off.

The more recent tendency towards longer locomotive runs to increase the mileage run per locomotive, thus reducing the out-of-service time and reducing the number of locomotives required for the service, has intensified the demand for an automatic oil delivering device of capacity to ensure sufficient lubrication the entire distance without depending on the whims of the different engineers as to how much oil is required to properly lubricate.

An automatic oil delivering device was also thought to be particularly desirable and economical on switching locomotives run by two or three crews in the twenty-four hours, in that it would deliver a definite quantity of oil only when the locomotive was working, so that one crew could not blame the other for using the entire twenty-four hours supply in eight hours.

The task set out to accomplish was to build an instrument that would deliver a predetermined amount of oil into the steam every revolution of the drivers, with nothing to turn on or off before or after filling, or before or after running, summer or winter in any climate; in other words one that was absolutely automatic in operation.

The Schlacks Force Feed Lubricator

The Schlacks force feed lubricator is attached to a bracket on the back head of either valve chamber.

A lever, that is actuated by the locomotive valve gear, works a ratchet that turns a cam shaft.

The cam is made so as to impart a quick suction stroke to the plungers which quickly fills the pump chambers with oil that is delivered to the engine in a series of successive short return strokes, completing the cycle.

Tests demonstrated that, notwithstanding the fact that the lubricator was delivering oil into the pipes at pracically every revolution of the drivers, the oil was not being delivered into the steam with the same regularity.

It was also found that the vacuum formed in the valves and cylinders while drifting and by the reduction in cylinder casting temperatures when standing still, would cause more oil to be drawn from the reservoir than the pumps delivered.

Considerable difficulty also was encountered due to condensation passing by the ordinary terminal check valve and gravitating to the bottom of the oil reservoir, where the pumps would take it up and deliver the water, instead of the oil, back into the stream; in fact, in some instances the rapid accumulation of the water in the oil reservoir would float almost all of the oil out; being displaced by the water.

It is an axiom that the oil pipe pressure at the delivery end in any form of lubrication must be in excess of the steam pressure into which the oil is to be injected.

On account of the fluctuating steam pipe pressures in a locomotive (sometimes running with full throttle and at other times drifting), it was early realized that a valve must be provided at the delivery end of each oil pipe that would maintain a constant oil pipe pressure in excess of the maximum working steam pressure, to avoid the engines running dry while the lubricator pump was building up an oil pipe pressure in excess of the steam pipe pressure.

These problems were solved by the design of a diaphragm terminal check valve, which:—

1. Maintains a constant oil pipe pressure.
2. Will not let condensation gravitate to the lubricator, and
3. Will not let a vacuum draw oil out of the pipe.

In other words, this valve permits a duplication of the lubricator pump's action at the delivery end of the oil pipe, irrespective of the length of the oil pipe or the steam pressure fluctuations into which the oil is being forced, and, in combination with the quick suction stroke plungers, ensures an injection of oil into the steam every revolution of the drivers.

Valve and Cylinder Lubrication of Superheater Locomotives

The hydrostatic lubricator oil pipe pressure at the point of delivery into the steam depends on the boiler steam pressure in the lubricator itself, plus the water-pressure-head between the lubricator and the point of delivery into the steam pipe. The difference in pressure into which the oil must be delivered and the hydrostatic-lubricator-oil-pipe-pressure at the point of delivery on modern power is reduced:

1. By reduced difference in height of the lubricator and the point of delivery into the steam, due to raising the point of delivery on account of the increased diameter of cylinders and the piston valve chamber on top, and the impossibility of increasing the height of the lubricator on account of the limited right of way overhead clearance.

2. By the increased oil pipe friction due to its increased length account of the greater distance from the lubricator in the cab to the steam pipe of the steam.

It has been proved that better cylinder lubrication can be obtained by injecting the oil a sufficient distance, twenty or more inches, above the valve to enable the oil to mix thoroughly with the steam before it reaches the valve, thus insuring a smaller proportion of oil passing out to exhaust by leaking valve packing rings. If, therefore, the point of delivery into the steam of the hydrostatic lubricator is made high up in the steam pipe, to get the benefit of thoroughly mixing the oil in all the steam, the difference in height from the point of delivery and the lubricator itself is so reduced as to make dependable, regular delivery impossible on long continuous runs with a fixed throttle. And, again, if the point of delivery into the steam is lowered to increase the oil pipe pressure by increasing the difference in height from it to the lubricator, the oil has not sufficient time to thoroughly mix with all the steam.

This has caused resort to different kinds of atomizing devices.

The reasons for adopting the force-feed lubricator, aside from the above facts, are:

1. It eliminates the human element.
2. It relieves the engineman of the duty of adjusting the feeds, which permits him that much more time for looking ahead.
3. It permits of official control of oil consumption.
4. It removes one more item from the already crowded cab.
5. It contains no explosive pressure.
6. It can be more readily filled.
7. It lends itself well to extended locomotive runs, which have been recently carried to limits, especially with the oil burning locomotive, heretofore thought impossible.
8. There is nothing to turn on or off before or after filling, before or after running, summer or winter, in any climate.
9. With ordinary careful handling, cleaning or draining oftener than when the engine is shopped is unnecessary.
10. The practically unlimited pressure into which the force feed lubricator can inject the oil.
11. The reduction of carbon incrustation due to delivering the lubricant only when the locomotive is running.
12. The increase in miles run to a pint of oil and to a set of valve and cylinder packing as well as valve stem and piston rod packing.

There are so many things besides the lubricator itself to be taken in consideration, to obtain the best results of valve and cylinder lubrication of superheater locomotives, that I believe it unwise to make a test of lubricator without, at the same time, arriving at a conclusion, from exhaustive tests, regarding all the contributing factors.

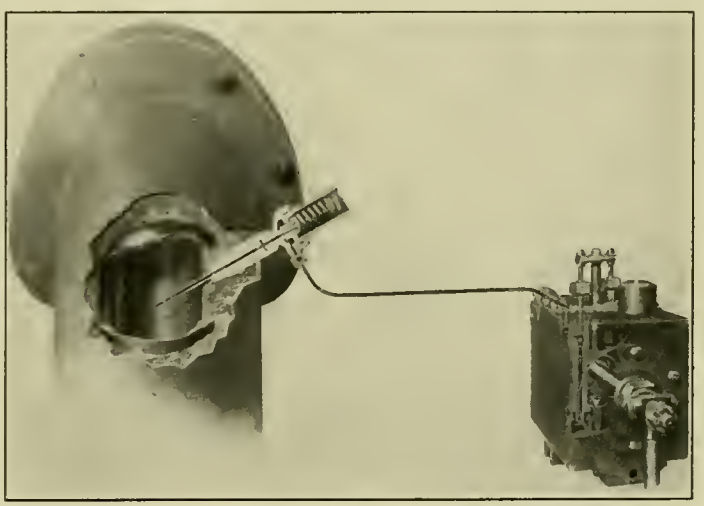
The lubricator is merely a lubricant delivering instrument, and the best is the one that will most consistently deliver a predetermined quantity of the best lubricant for the purpose into the steam, every revolution of the drivers, through the greatest variation of speeds and atmospheric temperatures, for the longest time, with the least attention.

Consistent with the above then, the best lubricant should be determined upon and used. Thorough investigation should determine the best place to inject the best lubricant by the best lubricator.

The above should be used together with the best form and material of valve and cylinder packing and bushings made to accurate dimensions and alignment. It is impossible to lubricate a locomotive whose valve and cylinder packing will not pack.

The best form of drifting device should be determined upon; one that is absolutely automatic that will not propel the locomotive and contains the least number of working parts requiring attention. Such a device would make relief or shifting valves unnecessary and even undesirable and would permit of the use of vacuum closing cylinder cocks. Such a device should avoid drawing front end cinders and gases into the valves and cylinders when drifting, which combined with excessive oil delivery, when not required, is the cause of carbon incrustation.

Difficulty in lubricating certain locomotives has been experienced until it was discovered that the gauge cocks



Illustrating Movement of Oil From Reservoir Into Steam Pipe of Locomotive, Schlacks System Force Feed Lubrication

and water glasses were located too high above the crown sheet thus causing working water on account of a too restricted steam space.

The greatest number of two feeds force feed lubricators in use are bolted to a one inch thick wrought iron bracket which is attached to the back head of either left or right valve chamber by three long valve head studs.

The lubricator lever is connected to a driving rod which is actuated by a stud connection on an extension of the locomotive valve gear combination lever.

The oil is conveyed through one-quarter inch by three-eighths inch annealed copper tubes from the lubricator to diaphragm terminal check valves located as high as possible in both the left and right outside steam pipes. The check valves are set to maintain oil pipe pressure of three hundred pounds per square inch.

The check valves are screwed into steam pipe plugs, which at an angle of 30 degrees from the horizontal, convey the oil to the center line of the steam pipe.

A one-quarter by three-eighths annealed copper tube run against the boiler under the lagging from an independent connection to the fountain near the cab, conveys steam to a heater chamber under the lubricator for heating the oil in the reservoir. This heater line is never regulated.

A three-eighths inch black iron pipe drains the heater chamber into the cylinder exhaust channel, or into the air pump exhaust.